Kinematic Analysis Report

Auto-Solver

February 3, 2026

1 Step 1: DH Matrices

The DH transformation matrices:

Matrix T_1

$$\begin{bmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 & L\cos(\theta_1) \\ \sin(\theta_1) & \cos(\theta_1) & 0 & L\sin(\theta_1) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix T_2

$$\begin{bmatrix} \cos(\theta_2) & 0 & -\sin(\theta_2) & 2L\cos(\theta_2) \\ \sin(\theta_2) & 0 & \cos(\theta_2) & 2L\sin(\theta_2) \\ 0 & -1 & 0 & 5L \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix T_3

$$\begin{bmatrix} -\cos(\theta_3) & 0 & -\sin(\theta_3) & 0 \\ -\sin(\theta_3) & 0 & \cos(\theta_3) & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix T_4

$$\begin{bmatrix} \cos(\theta_4) & 0 & -\sin(\theta_4) & \frac{5L\cos(\theta_4)}{2} \\ \sin(\theta_4) & 0 & \cos(\theta_4) & \frac{5L\sin(\theta_4)}{2} \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix T_5

$$\begin{bmatrix} \cos(\theta_5) & 0 & \sin(\theta_5) & \frac{5L\cos(\theta_5)}{2} \\ \sin(\theta_5) & 0 & -\cos(\theta_5) & \frac{5L\sin(\theta_5)}{2} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix T_6

$$\begin{bmatrix} -\cos(\theta_6) & 0 & \sin(\theta_6) & 0 \\ -\sin(\theta_6) & 0 & -\cos(\theta_6) & 0 \\ 0 & -1 & 0 & \frac{5L}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix T_7

$$\begin{bmatrix} \cos(\theta_7) & 0 & -\sin(\theta_7) & 2L\cos(\theta_7) \\ \sin(\theta_7) & 0 & \cos(\theta_7) & 2L\sin(\theta_7) \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2 Step 2: Loop Equation

$$T_1T_2T_3T_4T_5T_6T_7 = I$$

3 Step 3: Permutations

Set 1: $T_1T_2T_3T_4 = T_7^{-1}T_6^{-1}T_5^{-1}$

Set 2: $T_2T_3T_4T_5 = T_1^{-1}T_7^{-1}T_6^{-1}$

Set 3: $T_3T_4T_5T_6 = T_2^{-1}T_1^{-1}T_7^{-1}$

Set 4: $T_4T_5T_6T_7 = T_3^{-1}T_2^{-1}T_1^{-1}$

Set 5: $T_5T_6T_7T_1 = T_4^{-1}T_3^{-1}T_2^{-1}$

Set 6: $T_6T_7T_1T_2 = T_5^{-1}T_4^{-1}T_3^{-1}$

Set 7: $T_7T_1T_2T_3 = T_6^{-1}T_5^{-1}T_4^{-1}$

4 Step 4: Simplified Equations

Set 1

Set 1, (p˙x)
$$: \frac{L\left(-5\sin(\theta_4)\sin(\theta_1+\theta_2) - 5\sin(\theta_7) + 2\cos(\theta_1) - 5\cos(\theta_3)\cos(\theta_4)\cos(\theta_1+\theta_2) - 5\cos(\theta_6)\cos(\theta_7) + 4\cos(\theta_1+\theta_2) + 4\right)}{2}$$
$$= 0$$

Set 1, (o˙y) : $\sin(\theta_1)\sin(\theta_3)\cos(\theta_2) + \sin(\theta_2)\sin(\theta_3)\cos(\theta_1) + \sin(\theta_5)\sin(\theta_6)$
$$= 0$$

Set 1, (a˙y) : $-\sin(\theta_1)\sin(\theta_2)\cos(\theta_4) + \sin(\theta_1)\sin(\theta_4)\cos(\theta_2)\cos(\theta_3) + \sin(\theta_2)\sin(\theta_4)\cos(\theta_1)\cos(\theta_3)$
$$+ \cos(\theta_1)\cos(\theta_2)\cos(\theta_4) - \cos(\theta_6)$$
$$= 0$$

Set 1, (p˙y) : $\frac{L\left(2\sin(\theta_1) + 5\sin(\theta_4)\cos(\theta_1+\theta_2) - 5\sin(\theta_6) - 5\sin(\theta_1+\theta_2)\cos(\theta_3)\cos(\theta_4) + 4\sin(\theta_1+\theta_2)\right)}{2}$
$$= 0$$

$$\text{Set 1, (o˙z)} : \frac{\sin(\theta_5)\sin^2(\theta_6)\sin(\theta_7)}{\cos(\theta_6)} - \frac{\sin(\theta_5)\sin(\theta_7)}{\cos(\theta_6)} + \cos(\theta_3) - \cos(\theta_5)\cos(\theta_7)$$
$$= 0$$
$$\text{Set 1, (a˙z)} : -\sin(\theta_3)\sin(\theta_4) - \sin(\theta_6)\sin(\theta_7)$$
$$= 0$$
$$\text{Set 1, (p˙z)} : \frac{5L(\sin(\theta_3)\cos(\theta_4) + \sin(\theta_7)\cos(\theta_6) - \cos(\theta_7) + 2)}{2}$$
$$= 0$$

Set 2

$$\text{Set 2, (a˙x)} : -\frac{\sin^2(\theta_1)\sin(\theta_7)}{\cos(\theta_1)} - \sin(\theta_2)\sin(\theta_4)\sin(\theta_5) - \sin(\theta_3)\cos(\theta_2)\cos(\theta_5)$$
$$- \sin(\theta_5)\cos(\theta_2)\cos(\theta_3)\cos(\theta_4) + \frac{\sin(\theta_7)}{\cos(\theta_1)}$$
$$= 0$$
$$\text{Set 2, (o˙y)} : \frac{\sin(\theta_1)\sin(\theta_6)\sin^2(\theta_7)}{\cos(\theta_7)} - \frac{\sin(\theta_1)\sin(\theta_6)}{\cos(\theta_7)} + \sin(\theta_2)\sin(\theta_4)\cos(\theta_3)$$
$$- \cos(\theta_1)\cos(\theta_6) + \cos(\theta_2)\cos(\theta_4)$$
$$= 0$$
$$\text{Set 2, (a˙y)} : -\sin(\theta_1)\sin(\theta_7) - \sin(\theta_2)\sin(\theta_3)\cos(\theta_5) - \sin(\theta_2)\sin(\theta_5)\cos(\theta_3)\cos(\theta_4) + \sin(\theta_4)\sin(\theta_5)\cos(\theta_2)$$
$$= 0$$
$$\text{Set 2, (n˙z)} : \sin(\theta_3)\cos(\theta_4)\cos(\theta_5) + \sin(\theta_5)\cos(\theta_3) + \frac{\sin^2(\theta_6)\sin(\theta_7)}{\cos(\theta_6)} - \frac{\sin(\theta_7)}{\cos(\theta_6)}$$
$$= 0$$
$$\text{Set 2, (o˙z)} : -\sin(\theta_3)\sin(\theta_4) - \sin(\theta_6)\sin(\theta_7)$$
$$= 0$$
$$\text{Set 2, (a˙z)} : \sin(\theta_3)\sin(\theta_5)\cos(\theta_4) - \cos(\theta_3)\cos(\theta_5) + \cos(\theta_7)$$
$$= 0$$
$$\text{Set 2, (p˙z)} : \frac{5L(\sin(\theta_3)\cos(\theta_4)\cos(\theta_5) + \sin(\theta_3)\cos(\theta_4) + \sin(\theta_5)\cos(\theta_3) - \cos(\theta_7) + 2)}{2}$$
$$= 0$$

Set 3

$$\text{Set 3, (p˙x)}$$
$$: \frac{L\left(-4\sin(\theta_1)\sin(\theta_2) - 5\sqrt{2}\sin(\theta_3)\cos\left(\theta_5 + \frac{\pi}{4}\right) - 5\sqrt{2}\sin\left(\theta_5 + \frac{\pi}{4}\right)\cos(\theta_3)\cos(\theta_4) + 4\cos(\theta_1)\cos(\theta_2) + 2\cos(\theta_2) - 5\cos(\theta\right.}{2}$$
$$= 0$$
$$\text{Set 3, (n˙y)} : -\sin(\theta_3)\sin(\theta_4)\sin(\theta_6) + \sin(\theta_3)\cos(\theta_4)\cos(\theta_5)\cos(\theta_6) + \sin(\theta_5)\cos(\theta_3)\cos(\theta_6) - \sin(\theta_7)$$
$$= 0$$
$$\text{Set 3, (o˙y)} : \sin(\theta_3)\sin(\theta_5)\cos(\theta_4) - \cos(\theta_3)\cos(\theta_5) + \cos(\theta_7)$$
$$= 0$$
$$\text{Set 3, (a˙y)} : -\sin(\theta_3)\sin(\theta_4)\cos(\theta_6) - \sin(\theta_3)\sin(\theta_6)\cos(\theta_4)\cos(\theta_5) - \sin(\theta_5)\sin(\theta_6)\cos(\theta_3)$$
$$= 0$$
$$\text{Set 3, (p˙y)} : \frac{5L\left(-\sqrt{2}\sin(\theta_3)\sin\left(\theta_5 + \frac{\pi}{4}\right)\cos(\theta_4) - \sin(\theta_3)\cos(\theta_4) + \sqrt{2}\cos(\theta_3)\cos\left(\theta_5 + \frac{\pi}{4}\right) - 2\right)}{2}$$
$$= 0$$
$$\text{Set 3, (o˙z)} : -\frac{\sin^2(\theta_1)\sin(\theta_2)\sin(\theta_7)}{\cos(\theta_1)} + \sin(\theta_1)\sin(\theta_7)\cos(\theta_2) + \frac{\sin(\theta_2)\sin(\theta_7)}{\cos(\theta_1)} - \sin(\theta_4)\sin(\theta_5)$$
$$= 0$$
$$\text{Set 3, (a˙z)} : -\sin(\theta_1)\sin(\theta_2) + \sin(\theta_4)\sin(\theta_6)\cos(\theta_5) + \cos(\theta_1)\cos(\theta_2) - \cos(\theta_4)\cos(\theta_6)$$
$$= 0$$
$$\text{Set 3, (p˙z)} : \frac{L\left(-4\sin(\theta_1)\cos(\theta_2) - 4\sin(\theta_2)\cos(\theta_1) - 2\sin(\theta_2) + 5\sqrt{2}\sin(\theta_4)\sin\left(\theta_5 + \frac{\pi}{4}\right) + 5\sin(\theta_4)\right)}{2}$$
$$= 0$$

Set 4

$$\begin{aligned}
&\text{Set 4, (a˙x)} : -\sin(\theta_3) - \sin(\theta_4)\sin(\theta_6)\sin(\theta_7) - \sin(\theta_5)\cos(\theta_4)\cos(\theta_7) + \sin(\theta_7)\cos(\theta_4)\cos(\theta_5)\cos(\theta_6)\\
&\qquad = 0\\
&\text{Set 4, (o˙y)} : \sin(\theta_1)\sin(\theta_2) - \sin(\theta_4)\sin(\theta_6)\cos(\theta_5) - \cos(\theta_1)\cos(\theta_2) + \cos(\theta_4)\cos(\theta_6)\\
&\qquad = 0\\
&\text{Set 4, (a˙y)} : -\sin(\theta_4)\sin(\theta_5)\cos(\theta_7) + \sin(\theta_4)\sin(\theta_7)\cos(\theta_5)\cos(\theta_6) + \sin(\theta_6)\sin(\theta_7)\cos(\theta_4)\\
&\qquad = 0\\
&\text{Set 4, (o˙z)} : -\frac{\sin(\theta_1)\sin^2(\theta_2)\sin(\theta_3)}{\cos(\theta_2)} + \frac{\sin(\theta_1)\sin(\theta_3)}{\cos(\theta_2)} + \sin(\theta_2)\sin(\theta_3)\cos(\theta_1) + \sin(\theta_5)\sin(\theta_6)\\
&\qquad = 0\\
&\text{Set 4, (a˙z)} : -\sin(\theta_5)\sin(\theta_7)\cos(\theta_6) + \cos(\theta_3) - \cos(\theta_5)\cos(\theta_7)\\
&\qquad = 0\\
&\text{Set 4, (p˙z)}\\
&: \frac{L\left(-2\sin(\theta_3)\cos(\theta_2) - 4\sin(\theta_3) + 4\sin(\theta_5)\cos(\theta_6)\cos(\theta_7) - 4\sin(\theta_7)\cos(\theta_5) - 10\cos(\theta_3) + 5\sqrt{2}\cos\left(\theta_5 + \frac{\pi}{4}\right)\right)}{2}\\
&= 0
\end{aligned}$$

Set 5

$$\begin{aligned}
&\text{Set 5, (a˙x)} : \frac{\sin(\theta_3)\sin^2(\theta_4)}{\cos(\theta_4)} - \frac{\sin(\theta_3)}{\cos(\theta_4)} - \sin(\theta_5)\cos(\theta_7) + \sin(\theta_7)\cos(\theta_5)\cos(\theta_6)\\
&\qquad = 0\\
&\text{Set 5, (n˙y)} : -\sin(\theta_1)\sin(\theta_5)\sin(\theta_6) + \frac{\sin^2(\theta_2)\sin(\theta_3)}{\cos(\theta_2)} - \frac{\sin(\theta_3)}{\cos(\theta_2)}\\
&\qquad\quad - \sin(\theta_5)\cos(\theta_1)\cos(\theta_6)\cos(\theta_7) + \sin(\theta_7)\cos(\theta_1)\cos(\theta_5)\\
&\qquad = 0\\
&\text{Set 5, (o˙y)} : \sin(\theta_1)\sin(\theta_5)\cos(\theta_6)\cos(\theta_7) - \sin(\theta_1)\sin(\theta_7)\cos(\theta_5) - \sin(\theta_2)\sin(\theta_3) - \sin(\theta_5)\sin(\theta_6)\cos(\theta_1)\\
&\qquad = 0\\
&\text{Set 5, (a˙y)} : \sin(\theta_5)\sin(\theta_7)\cos(\theta_6) - \cos(\theta_3) + \cos(\theta_5)\cos(\theta_7)\\
&\qquad = 0\\
&\text{Set 5, (o˙z)} : \sin(\theta_1)\sin(\theta_6)\cos(\theta_7) + \frac{\sin(\theta_2)\sin^2(\theta_3)\sin(\theta_4)}{\cos(\theta_3)} - \frac{\sin(\theta_2)\sin(\theta_4)}{\cos(\theta_3)}\\
&\qquad\quad + \cos(\theta_1)\cos(\theta_6) - \cos(\theta_2)\cos(\theta_4)\\
&\qquad = 0\\
&\text{Set 5, (a˙z)} : \sin(\theta_3)\sin(\theta_4) + \sin(\theta_6)\sin(\theta_7)\\
&\qquad = 0\\
&\text{Set 5, (p˙z)} : L\left(\sin(\theta_1)\cos(\theta_6) - 5\sin(\theta_3)\sin(\theta_4) + 2\sin(\theta_4)\cos(\theta_3) - \sin(\theta_6)\cos(\theta_1)\cos(\theta_7) - 2\sin(\theta_6)\cos(\theta_7)\right)\\
&\qquad = 0
\end{aligned}$$

Set 6

$$\begin{aligned}
&\text{Set 6, (o˙y)} : -\sin(\theta_3)\sin(\theta_4) - \sin(\theta_6)\sin(\theta_7)\\
&\qquad = 0\\
&\text{Set 6, (a˙y)} : -\sin(\theta_1)\sin(\theta_2)\cos(\theta_6) + \sin(\theta_1)\sin(\theta_6)\cos(\theta_2)\cos(\theta_7) + \sin(\theta_2)\sin(\theta_6)\cos(\theta_1)\cos(\theta_7)\\
&\qquad\quad + \cos(\theta_1)\cos(\theta_2)\cos(\theta_6) - \cos(\theta_4)\\
&\qquad = 0\\
&\text{Set 6, (p˙y)} : L\left(\sin(\theta_1)\cos(\theta_6) + 5\sin(\theta_6)\sin(\theta_7) - \sin(\theta_6)\cos(\theta_1)\cos(\theta_7) - 2\sin(\theta_6)\cos(\theta_7)\cos(\theta_1+\theta_2)\right.\\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\left. - 2\sin(\theta_6)\cos(\theta_7) + 2\sin(\theta_1+\theta_2)\cos(\theta_6)\right)\\
&\qquad = 0\\
&\text{Set 6, (o˙z)} : -\frac{\sin(\theta_3)\sin^2(\theta_4)\sin(\theta_5)}{\cos(\theta_4)} + \frac{\sin(\theta_3)\sin(\theta_5)}{\cos(\theta_4)} - \cos(\theta_3)\cos(\theta_5) + \cos(\theta_7)\\
&\qquad = 0
\end{aligned}$$

$$\text{Set 6, (a˙z)} : \sin(\theta_1)\sin(\theta_7)\cos(\theta_2) + \sin(\theta_2)\sin(\theta_7)\cos(\theta_1) - \sin(\theta_4)\sin(\theta_5) = 0$$

$$\text{Set 6, (p˙z)} : \frac{L\left(5\sin(\theta_5) - 2\sin(\theta_7)\cos(\theta_1) - 4\sin(\theta_7)\cos(\theta_1+\theta_2) - 4\sin(\theta_7) - 10\cos(\theta_7) + 5\right)}{2} = 0$$

Set 7

$$\text{Set 7, (a˙x)} : \sin(\theta_1)\sin(\theta_2)\sin(\theta_3)\cos(\theta_7) - \sin(\theta_3)\cos(\theta_1)\cos(\theta_2)\cos(\theta_7) + \frac{\sin(\theta_5)\sin^2(\theta_6)}{\cos(\theta_6)} - \frac{\sin(\theta_5)}{\cos(\theta_6)} + \sin(\theta_7)\cos(\theta_3) = 0$$

$$\text{Set 7, (p˙x)} : \frac{L\left(-10\sin(\theta_7) + 2\cos(\theta_1)\cos(\theta_7) - 5\cos(\theta_5)\cos(\theta_6) - 5\cos(\theta_6) + 4\cos(\theta_7)\cos(\theta_1+\theta_2) + 4\cos(\theta_7)\right)}{2} = 0$$

$$\text{Set 7, (n˙y)} : \sin(\theta_1)\sin(\theta_2)\sin(\theta_7)\cos(\theta_3) + \sin(\theta_3)\cos(\theta_7) - \frac{\sin^2(\theta_4)\sin(\theta_5)}{\cos(\theta_4)} + \frac{\sin(\theta_5)}{\cos(\theta_4)} - \sin(\theta_7)\cos(\theta_1)\cos(\theta_2)\cos(\theta_3) = 0$$

$$\text{Set 7, (o˙y)} : -\sin(\theta_1)\sin(\theta_7)\cos(\theta_2) - \sin(\theta_2)\sin(\theta_7)\cos(\theta_1) + \sin(\theta_4)\sin(\theta_5) = 0$$

$$\text{Set 7, (a˙y)} : \sin(\theta_1)\sin(\theta_2)\sin(\theta_3)\sin(\theta_7) - \sin(\theta_3)\sin(\theta_7)\cos(\theta_1)\cos(\theta_2) - \cos(\theta_3)\cos(\theta_7) + \cos(\theta_5) = 0$$

$$\text{Set 7, (p˙y)} : \frac{L\left(-5\sin(\theta_5) + 2\sin(\theta_7)\cos(\theta_1) + 4\sin(\theta_7)\cos(\theta_1+\theta_2) + 4\sin(\theta_7) + 10\cos(\theta_7) - 5\right)}{2} = 0$$

$$\text{Set 7, (o˙z)} : \sin(\theta_1)\sin(\theta_2) + \frac{\sin(\theta_4)\sin^2(\theta_5)\sin(\theta_6)}{\cos(\theta_5)} - \frac{\sin(\theta_4)\sin(\theta_6)}{\cos(\theta_5)} - \cos(\theta_1)\cos(\theta_2) + \cos(\theta_4)\cos(\theta_6) = 0$$

$$\text{Set 7, (a˙z)} : \sin(\theta_1)\sin(\theta_3)\cos(\theta_2) + \sin(\theta_2)\sin(\theta_3)\cos(\theta_1) + \sin(\theta_5)\sin(\theta_6) = 0$$

$$\text{Set 7, (p˙z)} : \frac{L\left(-2\sin(\theta_1) + 5\sin(\theta_6)\cos(\theta_5) + 5\sin(\theta_6) - 4\sin(\theta_1+\theta_2)\right)}{2} = 0$$

5 Step 5: Single Driver Analysis

No single driver solution found.

6 Step 6: Dual Driver Analysis

6.1 Inputs: θ_3, θ_5

1. Solve θ_4 using Set 3, (p˙y)
   Form: $A\sin(\theta_4) + B\cos(\theta_4) + C = 0$
   Poly: $(C-B)t^2 + 2At + (B+C) = 0, \quad t = \tan(\theta_4/2)$
   Branches: $t = \frac{-2A \pm \sqrt{4A^2 - 4(C^2-B^2)}}{2(C-B)}$

$$A = 0$$
$$B = 0$$
$$C = \frac{5L\left(\sqrt{2}\cos(\theta_3)\cos\left(\theta_5 + \frac{\pi}{4}\right) - 2\right)}{2}$$

2. Solve θ_7 using Set 2, (a˙z)
   Form: $A\sin(\theta_7) + B\cos(\theta_7) + C = 0$
   Poly: $(C-B)t^2 + 2At + (B+C) = 0, \quad t = \tan(\theta_7/2)$
   Branches: $t = \frac{-2A \pm \sqrt{4A^2 - 4(C^2 - B^2)}}{2(C-B)}$

$$A = 0$$
$$B = 1$$
$$C = \sin(\theta_3)\sin(\theta_5)\cos(\theta_4) - \cos(\theta_3)\cos(\theta_5)$$

3. Solve θ_6 using Set 1, (n˙z)
   Form: $A\sin(\theta_6) + B\cos(\theta_6) + C = 0$
   Poly: $(C-B)t^2 + 2At + (B+C) = 0, \quad t = \tan(\theta_6/2)$
   Branches: $t = \frac{-2A \pm \sqrt{4A^2 - 4(C^2 - B^2)}}{2(C-B)}$

$$A = 0$$
$$B = 0$$
$$C = \tilde{\infty}\left(\frac{\sin^2(\theta_5)\sin(\theta_7)}{\cos(\theta_5)} - \frac{\sin(\theta_7)}{\cos(\theta_5)}\right) + \sin(\theta_3)\cos(\theta_4) + \sin(\theta_5)\cos(\theta_7)$$

4. Solve θ_2 using Set 4, (p˙x)
   Form: $A\sin(\theta_2) + B\cos(\theta_2) + C = 0$
   Poly: $(C-B)t^2 + 2At + (B+C) = 0, \quad t = \tan(\theta_2/2)$
   Branches: $t = \frac{-2A \pm \sqrt{4A^2 - 4(C^2 - B^2)}}{2(C-B)}$

$$A = 0$$
$$B = 0$$
$$C = \frac{L\left(10\sin(\theta_3) + 4\sin(\theta_4)\sin(\theta_6)\cos(\theta_7) - 4\sin(\theta_5)\sin(\theta_7)\cos(\theta_4) + 5\sqrt{2}\sin\left(\theta_5 + \frac{\pi}{4}\right)\cos(\theta_4) - 4\cos(\theta_3) - 4\cos(\theta_4)\cos(\theta_5)\cos(\theta_6)\cos(\theta_7) +\right.}{2}$$

5. Solve θ_1 using Set 1, (n˙x)
   Form: $A\sin(\theta_1) + B\cos(\theta_1) + C = 0$
   Poly: $(C-B)t^2 + 2At + (B+C) = 0, \quad t = \tan(\theta_1/2)$
   Branches: $t = \frac{-2A \pm \sqrt{4A^2 - 4(C^2 - B^2)}}{2(C-B)}$

$$A = \sin(\theta_2)\cos(\theta_3)\cos(\theta_4) - \sin(\theta_4)\cos(\theta_2)$$
$$B = -\sin(\theta_2)\sin(\theta_4) - \cos(\theta_2)\cos(\theta_3)\cos(\theta_4)$$
$$C = -\frac{\sin^2(\theta_5)\sin^2(\theta_6)\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \frac{\sin^2(\theta_5)\sin^2(\theta_6)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \frac{\sin^2(\theta_5)\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)}$$
$$- \frac{\sin^2(\theta_5)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \sin(\theta_5)\sin(\theta_7) + \frac{\sin^2(\theta_6)\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)}$$
$$- \frac{\sin^2(\theta_6)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} - \frac{\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \frac{1}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)}$$

6.2 Inputs: θ_4, θ_5

1. Solve θ_3 using Set 3, (p˙y)
   Form: $A\sin(\theta_3) + B\cos(\theta_3) + C = 0$
   Poly: $(C-B)t^2 + 2At + (B+C) = 0, \quad t = \tan(\theta_3/2)$
   Branches: $t = \frac{-2A \pm \sqrt{4A^2 - 4(C^2 - B^2)}}{2(C-B)}$

$$A = 0$$
$$B = 0$$
$$C = -5L$$

2. Solve θ_7 using Set 2, (a˙z)
   Form: $A\sin(\theta_7) + B\cos(\theta_7) + C = 0$
   Poly: $(C-B)t^2 + 2At + (B+C) = 0, \quad t = \tan(\theta_7/2)$
   Branches: $t = \frac{-2A \pm \sqrt{4A^2 - 4(C^2 - B^2)}}{2(C-B)}$

$$A = 0$$
$$B = 1$$
$$C = \sin(\theta_3)\sin(\theta_5)\cos(\theta_4) - \cos(\theta_3)\cos(\theta_5)$$

3. Solve θ_6 using Set 1, (n˙z)
   Form: $A\sin(\theta_6) + B\cos(\theta_6) + C = 0$
   Poly: $(C-B)t^2 + 2At + (B+C) = 0, \quad t = \tan(\theta_6/2)$
   Branches: $t = \frac{-2A \pm \sqrt{4A^2 - 4(C^2 - B^2)}}{2(C-B)}$

$$A = 0$$
$$B = 0$$
$$C = \tilde{\infty}\left(\frac{\sin^2(\theta_5)\sin(\theta_7)}{\cos(\theta_5)} - \frac{\sin(\theta_7)}{\cos(\theta_5)}\right) + \sin(\theta_3)\cos(\theta_4) + \sin(\theta_5)\cos(\theta_7)$$

4. Solve θ_2 using Set 4, (p˙x)
   Form: $A\sin(\theta_2) + B\cos(\theta_2) + C = 0$
   Poly: $(C-B)t^2 + 2At + (B+C) = 0, \quad t = \tan(\theta_2/2)$
   Branches: $t = \frac{-2A \pm \sqrt{4A^2 - 4(C^2 - B^2)}}{2(C-B)}$

$$A = 0$$
$$B = 0$$
$$C = \frac{L\left(10\sin(\theta_3) + 4\sin(\theta_4)\sin(\theta_6)\cos(\theta_7) - 4\sin(\theta_5)\sin(\theta_7)\cos(\theta_4) + 5\sqrt{2}\sin\left(\theta_5 + \frac{\pi}{4}\right)\cos(\theta_4) - 4\cos(\theta_3) - 4\cos(\theta_4)\cos(\theta_5)\cos(\theta_6)\cos(\theta_7) + \right.}{2}$$

5. Solve θ_1 using Set 1, (n˙x)
   Form: $A\sin(\theta_1) + B\cos(\theta_1) + C = 0$
   Poly: $(C-B)t^2 + 2At + (B+C) = 0, \quad t = \tan(\theta_1/2)$
   Branches: $t = \frac{-2A \pm \sqrt{4A^2 - 4(C^2 - B^2)}}{2(C-B)}$

$$A = \sin(\theta_2)\cos(\theta_3)\cos(\theta_4) - \sin(\theta_4)\cos(\theta_2)$$
$$B = -\sin(\theta_2)\sin(\theta_4) - \cos(\theta_2)\cos(\theta_3)\cos(\theta_4)$$
$$C = -\frac{\sin^2(\theta_5)\sin^2(\theta_6)\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \frac{\sin^2(\theta_5)\sin^2(\theta_6)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \frac{\sin^2(\theta_5)\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)}$$
$$- \frac{\sin^2(\theta_5)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \sin(\theta_5)\sin(\theta_7) + \frac{\sin^2(\theta_6)\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)}$$
$$- \frac{\sin^2(\theta_6)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} - \frac{\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)} + \frac{1}{\cos(\theta_5)\cos(\theta_6)\cos(\theta_7)}$$

7 Step 7: Numerical Plots